

Networks and Markets

Final Project

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Part 1: Voting

1. In the American presidential elections, while the popular vote is used up to the state level, the electoral college decides the winner at the national level. Assuming there are only two candidates, is this system strategy-proof? Does it elect a Condorcet winner? Justify your answers. (Note: You can assume for simplicity that each state gets a single “vote” in a national election, and the state then runs an election by popular vote with however many people are in that state to determine which vote to cast at the national level. If you give a counter-example, you may choose freely the number of states and the number of voters in each state.)
2. Suppose we have an election between n candidates $c_1, \dots, c_n$, where each candidate is assigned a ranking (x_i, y_i) representing its stance on two political issues. Each voter has an ideal point (x, y) , and it ranks candidates based on the distance $\sqrt{(x - x_i)^2 + (y - y_i)^2}$, ranking candidates with a smaller distance higher. Is it always possible to pick a Condorcet winner? Prove or give a counterexample.
3. Suppose we want to run a popular vote election between two candidates A and B . There are 1,000,000 eligible voters and suppose 52% of them prefer A to B . Instead of running a full election where every person casts a vote, we poll m randomly selected people (with replacement, for simplicity) and ask them to report their preferred candidate. The result of the poll is the majority of the responses received.
 - (a) Compute a value of m so that the result of the poll is incorrect with probability at most 1%. (Use the Hoeffding/ Chernoff bounds, show your work.)
 - (b) Let n be the number of people in the population, ϵ be defined such that $(1/2 + \epsilon) \cdot n$ prefer A to B , and let δ be the desired accuracy (so the probability the result is incorrect is at most δ). Write your bound m as a function of n , ϵ , and δ .

If the number of people in the population increased by a factor of 10, how would that affect m ? If ϵ decrease by a factor of 2, how would that affect m ? If we want to increase our confidence by a factor of 10 ($\delta' = \delta/10$), how would that change m ? If $\epsilon = 1/n$ (so 1 person would be the deciding vote), what would this imply about m given your bound from above?
 - (c) In practice, what might be wrong with the above assumptions (i.e. why might we not use polls to run our elections)?

Part 2: Stable Matchings

4. For the following setting, find the female-optimal stable matching. Describe in detail how the Gale-Shapley algorithm arrives at the matching you find.

Females	Preferences	Males	Preferences
A	$X > W > Y > Z$	W	$D > B > C > A$
B	$X > W > Y > Z$	X	$D > B > A > C$
C	$X > W > Z > Y$	Y	$C > B > D > A$
D	$Y > W > Z > X$	Z	$D > B > C > A$

5. Prove that there exists a non-bipartite matching setting (where every individual has preferences over all other individuals) for which no stable matching exists.

Part 3: Wisdom of Crowds

6. In the setting described in Recitation 9 Slide 8, what happens if players choose sequentially, where Player 2 knows Player 1's choice, and every player i where $i > 2$ knows the choices of players $i - 1$ and $i - 2$? Are we in the "Wisdom" case where the majority is correct with high probability, or in the "Foolishness" case where there is substantial probability all choices are wrong? Assume players always choose their own signal when all else is equal. Explain your answer.

Part 4: PageRank and Social Networks

The objective of this question is to use the PageRank algorithm as a way to determine how “influential” a node is in a social network based on its in-links from influential nodes. For this question, we provided a template in python called ‘hw4.py’. You must use the template to submit your code, as we will grade your code in a (partially) automated way. *You should submit the hw4.py file, not a .pynb or other python file.*

7. Design an algorithm that runs the iterative ϵ -scaled PageRank algorithm for a specified number n of rounds on a given directed graph, with $\epsilon = 1/7$. Run it (with $n = 10$) on the examples from lecture 9 (15.1.left is slide 13, 15.1.right is slide 14, 15.2 is slide 15), as well as at least two other simple test cases with at least 10 nodes.
8. Now we’ll run PageRank on the Facebook data from “facebook_combined.txt”; remember that it has 4,039 nodes.
 - (a) Remember that this is an undirected graph! Before running your algorithms from the previous problem, implement a transformation into a directed graph, i.e. each undirected edge corresponds to two different directed edges. Include your code in hw4.py.
 - (b) Now, run the PageRank algorithms from the last problem on this new graph. You shouldn’t need n to be much higher than 10-20 for the algorithm to converge to a fixed point. Make sure to submit your code in the main function of hw4.py.
 - (c) Where did most of the score tend to end up in your experiments? Look at the nodes that have the highest or lowest scores; is there a consistent pattern among your trials? Include your analysis in the pdf write-up.
 - (d) Intuitively explain your results in terms of a measure of influence in a social network. Do you think that this is an accurate measurement? How could we try to improve it (for instance, by incorporating link strengths or other measures of popularity)? Include your answers in the pdf write-up.

Part 5: Semester Review

The following questions must be done *individually* with only the course notes, slides, or book as reference. Namely, **you are not allowed to collaborate with other students on this part of the assignment.**

9. Consider the following two player game between candidate 1 and 2 in a simplified 2-person election. There is an integer value V for winning the election, and each candidate i decides an integer price $p_i \geq 0$ to spend on advertising. The winner of the election i is the candidate who spends more on advertising, who gets utility $V - p_i$. The loser j gets utility $-p_j$. In the case of a tie when $p_1 = p_2$, the election is decided by a coin-toss, and we assume for simplicity that each candidate i gets utility $V/2 - p_i$.
 - (a) You find out that an election has been held and candidate 1 paid $p_1 > V$ on advertising. Show that there is no action profile where $p_1 > V$ is a best-response for candidate 1.
 - (b) In this next part, we are going to show that if $V \geq 3$, there are no PNE in this game. Recall that by part (a), we can immediately rule out any action profile where p_1 or p_2 are greater than V .

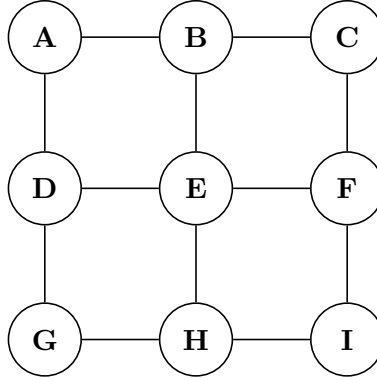


Figure 1: The layout of Chloe Cow County

- i. Prove that there are no PNE where $p_1 = p_2$. (Hint: Show that barely outspending your opponent always increases your utility.)
- ii. Now consider the case where $p_1 \neq p_2$, and suppose without loss of generality that $p_1 > p_2$. Show that there cannot be a PNE where $p_2 > 0$.
- iii. Finally, show that the action profile $(p_1, 0)$ cannot be a PNE.

Thus, there are no PNE where $p_1 = p_2$ or when $p_1 \neq p_2$. This covers everything, so we conclude there are no PNE when $V \geq 3$!

- (c) Compute the action profile that maximizes social welfare of the two candidates (justify why this maximizes social welfare). Describe how policy makers could put external pressures on the election to make the above outcome a PNE.

10. After graduating from Tel Aviv University, you find yourself consulting for Chloe Cow County. In the county, there are 9 towns connected by roads in a grid as depicted in Figure 1 (assume that the distance between two adjacent towns has length 1). The county is looking to open up schools in the area, and they need your help! Because of funding and student populations, each school must serve (exactly) two adjacent towns and each town can only send their students to a single school. A school will be placed either in one of the two towns it serves or somewhere along the road in between the two towns. In order to minimize travel time for its citizens, each town prefers that its school is as close to the town as possible with utility 1 minus the distance to its school (assuming it is assigned a school to begin with). Note that not every town will necessarily be assigned a school.

- (a) Encode the problem as an exchange network as a way to come up with a “stable” way to place schools in the county. We say that an assignment of schools is “stable” if no town can convince one of its neighboring towns to break away from the assignment and fund a new school with a different placement.

Specifically, give a graph G with edge values v , and explain how a stable outcome in G corresponds to a stable assignment of schools.

- (b) Ignoring stability, prove that valid placement of schools must leave at least one town without a school.

- (c) Find a stable outcome in this graph. You can compute the outcome either by hand or by using one of the algorithms outlined in the book. Explain why the outcome you found is stable.
- (d) **Bonus:** Prove that any county with towns laid out in an $n \times m$ grid (as in the 3×3 grid of Figure 1) will have a stable outcome. For which values of n, m will all towns have access to a school?
11. Ben Bitdiddle is designing sales strategies for Snowstorm Entertainment, in particular to convince people to play their new video game, ‘BOTA’, instead of the competing video game called ‘Age of Agents’. While doing market research, Ben realizes that people switch to BOTA if and only if more than a t_f fraction of their friends *AND* more than a t_c fraction of their coworkers have already switched to BOTA.

More formally, we model this social network with the tuple (V, E_f, E_c) , where each node in V is a person, and the edge sets E_f and E_c correspond to the friendship graph and coworking graph respectively. We say that a set $S \subseteq V$ is ‘jointly-cascading’ w.r.t. (V, E_f, E_c) and parameters t_f, t_c if, when every player in S starts by playing BOTA, running BRD restricted to $V \setminus S$ always ends with every player playing BOTA (where the best response is defined according to the switching rule above).

Figure 2 shows a specific friendship and coworking relationship graph between six people $V = \{0, \dots, 5\}$. Let E_c be the black dashed edges, and E_f be the red edges. Suppose $t_c = 0.49$ and $t_f = 0.65$. Let’s now analyze the ‘jointly-cascading’ behavior on Figure 2.

- (a) Find a single set $S \subseteq V$ s.t. $|S| = 2$ (i.e. has exactly two players) and S cascades w.r.t. (V, E_f) with adoption threshold t_f , and *ALSO* cascades w.r.t. (V, E_c) with adoption threshold t_c . (Note this is the regular notion of cascade – consider each graph separately.)
- (b) Does your choice of S above jointly-cascade w.r.t. (V, E_f, E_c) and the given parameters t_f, t_c ? Show your work.
- (c) Ben Bitdiddle is hoping to say that S jointly-cascades i.f.f. there does not exist $T \subseteq S \setminus V$ s.t. *EITHER* T has density $1 - t_f$ w.r.t. (V, E_f) *OR* T has density $1 - t_c$ w.r.t. (V, E_c) . Prove him wrong by showing a counterexample. (Hint: we’ve already done most of the work.)

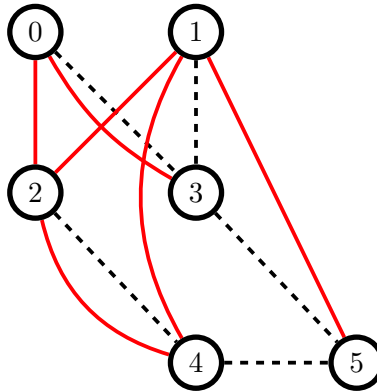


Figure 2: A friendship (red edges) and co-working (black dashed edges) graph

Part 6: Open-ended Project

This part can be submitted in triples or less. The PDF write-up for this part should not exceed a single page by much, and the code (if any) should be mostly based on code you already wrote for previous assignments. The goal of this assignment is to apply some topic from the course to a real world issue.

First, you should pick:

- (a) Some topical issue- anything that was in the news in the past 5 years should be fine, and you can ask us about a topic if you are not sure.
- (b) Some topic learned in class: a modeling technique or algorithm that you believe can be used to study said topical issue.

Then, you have two options:

- (a) Empirical:
 - i. Find a real-world data-set relevant to the issue you chose. Cite the source of your data. If no relevant data is publicly available you may generate “fake data”, as long as it seems reasonable. e.g., a market where all items have the same price is (usually) not realistic- some randomness should probably be involved.
 - ii. Explain how the data can be studied using the technique we learned in the course that you chose. Describe the modeling of the data as input for an algorithm we learned. The code you use for this should be included. It should mostly rely on code you already wrote for past assignments, but modifications are allowed- it will be tested manually.
 - iii. Run the experiment you described and say what conclusions one can draw. Describe your results. How might your results shed light on the real-world issue?
- (b) Theoretical:
 - i. Suggest a way to extend the technique or algorithm you chose to handle cases not handled by the technique or algorithm as shown in class. The extension should be non-trivial. Claims about correctness of algorithms you suggest should be justified.
 - ii. Explain how your suggestion helps handle the real-world problem you chose, and why this cannot be done (or at least is harder to do) using the technique or algorithm as shown in class.
 - iii. You may refer to books or published papers: you are not expected to present something that was never done before, just something that was not discussed in class. Cite any sources used.
 - iv. You may support your case with experiments (in which case the relevant code should be included), but you don’t have to.